



An Integrated Analytic Network Process-VIKOR Framework for Sustainability Indicator Prioritization and Glass Supplier Selection

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Abstract: Supplier selection has emerged as a critical strategic decision affecting both supply chain performance and long-term sustainability outcomes. Traditional selection methods have typically emphasized economic factors, often neglecting environmental and social considerations. This study develops an integrated analytic network process (ANP) and Višekriterijumska optimizacija i kompromisno rešenje (VIKOR; Multi-Criteria Optimization and Compromise Solution) framework for prioritizing sustainability indicators and selecting sustainable glass suppliers in Iran. Indicators were identified through literature review and expert consultation, structured across economic, environmental, and social dimensions. ANP determined indicator weights while accounting for interdependencies among criteria. These weights were then incorporated into VIKOR to rank suppliers and identify compromise solutions under conflicting objectives. The parameter v represented the emphasis on maximum group utility versus minimum individual regret. Results identified Qazvin Glass Company and Ardakan Yazd Glass Company as the top-performing suppliers. Sensitivity analysis confirmed ranking stability across parameter variations. By integrating interdependent sustainability indicators with compromise-based supplier ranking, the proposed ANP-VIKOR framework provides a structured decision-support approach through which economic, environmental, and social considerations can be incorporated simultaneously into supplier selection. The proposed framework offers a structured approach for integrating interdependent sustainability indicators into supplier selection and provides a transferable methodology for other manufacturing industries.

Keywords: Analytic network process technique; Multi-criteria optimization and compromise solution; Sustainable supply chain; Sustainable supplier selection; Glass industry

1 Introduction

Choosing the right supplier is among the most critical decisions in supply chain management, requiring evaluation of both measurable and subjective factors [1]. To address this complexity, researchers and practitioners have adopted various multi-criteria decision-making (MCDM) methods, including the Technique for Order Preference by Similarity to Ideal Solution (TOPSIS), the AHP, the analytic network process (ANP), Višekriterijumska optimizacija i kompromisno rešenje (VIKOR; Multi-Criteria Optimization and Compromise Solution), and fuzzy extensions—to handle supplier selection under uncertainty and sustainability constraints [2].

This investigation addresses three research questions: What are the effective sustainability indicators for supplier selection in the glass industry? How can glass industry suppliers be systematically prioritized? How does supplier prioritization impact supply chain sustainability outcomes? These questions are addressed through an integrated ANP-VIKOR framework, where the ANP captures interdependencies among sustainability criteria and VIKOR provides compromise solutions when conflicting criteria exist. The hybrid approach was selected because single-method applications cannot adequately capture the complexity of sustainability criteria interactions nor resolve conflicts between economic, environmental, and social objectives simultaneously.

Glass manufacturing plays a vital economic role both globally and within Iran. Iran is positioned among the world's ten largest glass-producing nations, with output surpassing 2 million tons annually [3]. This sector

directly employs more than 50,000 individuals and supplies numerous downstream industries, including construction, automotive, packaging, and consumer goods. Domestically, glass production accounts for roughly 2% of Iran's manufacturing gross domestic product (GDP) and is considered a strategic industry for reducing import dependence. From a sustainability perspective, the glass sector presents distinctive challenges compared to other manufacturing sectors:

- **Energy Intensity:** Glass production requires continuous furnace operation at temperatures reaching 1500 °C, making it one of the most energy-intensive manufacturing processes. This distinguishes it from textile or automotive assembly sectors where energy requirements are significantly lower [4].
- **Raw Material Dependence:** Glass manufacturing depends on specific raw materials (e.g., silica sand, soda ash, and limestone) whose quality variability directly affects product quality. Unlike electronics or automotive sectors where component specifications are standardized, raw material quality fluctuations present unique supplier selection challenges [5].
- **Logistical Considerations:** Glass raw materials are heavy and bulky, making transportation costs a significant factor in supplier selection. This creates distinct geographical proximity considerations not equally relevant in other sectors [6].
- **Environmental Regulations:** Glass furnace emissions (including nitrogen oxides, sulfur oxides, and particulate matter) face stringent environmental regulations, requiring suppliers to demonstrate environmental compliance [7].

Critical supplier-selection issues unique to the glass sector include silica sand quality consistency, uninterrupted supply for continuous furnace operation, transportation logistics optimization, and environmental compliance verification of mining operations [8].

The study is structured as follows: Section 2 presents the literature review, Section 3 describes the methodology, Section 4 presents the case study, Section 5 provides sensitivity analysis, Section 6 discusses findings, and Section 7 concludes with implications and future research directions.

2 Theoretical Background

Recent literature demonstrates increasing sophistication in sustainable supplier selection methodologies. This section reviews contemporary research emphasizing approaches published between 2021 and 2026, highlighting methodological innovations and empirical applications.

2.1 Recent Advances in Sustainable Supplier Selection

Afrasiabi et al. [9] developed a fuzzy hybrid MCDM approach for selecting sustainable and resilient suppliers. Their method integrated the fuzzy best-worst method with grey relational analysis and fuzzy TOPSIS. The study found that pollution control, environmental management systems, and risk awareness were the most influential criteria in manufacturing contexts. This research aligns with the emphasis of this current study on environmental criteria while extending consideration to supply chain resilience. Rodrigues et al. [10] applied a fuzzy decision-making trial and evaluation laboratory (DEMATEL) approach to identify and analyze sustainable supplier selection criteria in a glass packaging firm. Their findings showed expert consensus on criteria importance and interrelationships. The economic cluster had the strongest interactions, while the social cluster showed minimal influence. This pattern supports the limited role of social criteria observed in the present study, indicating industry-specific sustainability dynamics. Memari et al. [11] used intuitionistic fuzzy TOPSIS to evaluate sustainable suppliers in the automotive spare parts sector. Their model incorporated nine criteria and thirty sub-criteria. The approach produced a reliable sustainability ranking that was validated through a real-world case study. Li et al. [12] introduced a spherical fuzzy ranking method with two-step normalization to assess digital transformation capabilities in logistics firms. This work illustrates the trend toward higher-order fuzzy extensions in MCDM.

2.2 Hybrid Multi-Criteria Decision-Making Approaches

Chattopadhyay et al. [13] applied the D-MARCOS (D-numbers-based Measurement of Alternatives and Ranking according to COmpromise Solution) method to supplier selection in India's iron and steel sector, using D numbers to address decision uncertainty. Chen et al. [14] developed a rough-fuzzy DEMATEL-TOPSIS model for sustainable supplier selection in smart supply chains. The approach combined fuzzy sets for internal uncertainty with rough sets for external uncertainty. Nguyen et al. [15] combined fuzzy AHP with VIKOR for green supplier selection in Vietnam's coffee bean supply chain. Their analysis identified quantity reduction, solid waste generation, and logistics costs as the most critical criteria. The literature demonstrates clear trends toward hybrid frameworks combining multiple multi-criteria decision-making techniques to exploit complementary strengths [16]. Recent contributions by Kablan et al. [17] proposed a hybrid approach using interval-valued Fermatean fuzzy sets with step-wise weight assessment ratio analysis (SWARA) and weighted aggregated sum product assessment (WASPAS) for green supplier selection, advancing fuzzy extension applications. Badi et al. [16] created fuzzy MCDM models to evaluate barriers

in logistics outsourcing. Saha and Chatterjee [18] introduced a Fermatean fuzzy combined compromise solution (CoCoSo) method for selecting manufacturing vendors.

2.3 Glass Industry Applications

Rodrigues [19] used fuzzy DEMATEL-ANP for sustainable supplier selection in a Brazilian glass packaging firm. Economic criteria were most influential, while social criteria had minimal impact. Naeeni and Sabbaghi [20] designed a sustainable supply chain network for an Asian glass manufacturer, integrating environmental and social factors with economic objectives. Pourjavad and Mayorga [21] proposed a closed-loop supply chain optimization model for glass manufacturing, showing potential for cost savings and reduced environmental impact through forward-reverse logistics integration. Thanh [22] developed a dynamic decision support system for sustainable supplier selection under fuzzy conditions, applicable to glass industry contexts. The glass industry's distinct characteristics—including raw material extraction, energy-intensive production, and significant transportation requirements—demand specialized sustainability frameworks [23]. Liao et al. [24] used ANP + VIKOR for supplier selection in construction project procurement, demonstrating the combined method's effectiveness in complex decisions.

2.4 Identified Research Gaps

Despite substantial progress, several gaps persist in sustainable supplier selection literature. First, limited research specifically addresses the glass industry, despite its significant environmental footprint and raw material dependencies. Second, few studies integrate ANP and VIKOR for glass supplier selection, despite the ANP's ability to capture interdependencies and VIKOR's compromise solution advantages. Third, social sustainability criteria frequently receive insufficient attention, often remaining “inert” in decision models [19]. Finally, most studies treat sustainability dimensions independently rather than examining their interactions. This research addresses these gaps through an integrated ANP-VIKOR framework applied to Iran's glass industry. The criteria and sub-criteria summarized in Table 1 were derived from the reviewed literature.

Table 1. Supplier selection criteria

Dimensions	Criteria	Sub-Criteria	Description	Reference
Economic	Cost	Fixed price	Price provided by the supplier	[1, 15]
		Maintenance costs	Order receipt determines material usage and storage	[1, 15]
		After-sales service costs	Raw material inspection by supplier when issues arise	[1, 16]
	Quality	Quality	Raw material quality	[1]
		Internal material quality inspection process	Existence of the quality control system	[25]
	Delivery	Timely delivery	On-time delivery performance	[1, 15]
		Delay payment	Fine payment for late delivery	[21]
	Services	After-sales service	Service process and complaint resolution duration	[1]
	Flexibility	Flexibility in giving discounts	Discount for certain order quantities or liquidity	[1]
	Geographical location	–	Supplier proximity to the buyer, reducing transport costs	[1, 24]
Environmental	Environmental management system	International Organization for Standardization (ISO) 14001 certification	Environmental protection requirements	[11, 13]
	Resource consumption	Energy consumption	Energy use in raw material preparation	[1, 21]
	Green product	Dangerous gases	Hazardous gas production in the mining phase	[21, 24]
	Pollution reduction	Transportation	Vehicles used for raw material transport	[21, 24]
		Technology	Technology type for extraction and preparation	[21]
Social	Occupational health and safety	Provision of appropriate equipment in the workplace	Fire extinguishing systems and safety equipment	[21]
	Employee rights and welfare	Employment insurance	Social security coverage for workers	[14, 21]
		Standard working hours	Compliance with labor regulations	[1, 21]
	Disclosure of information	–	Information to customers and stakeholders	[1]
	Company brand	–	Supplier credit in local community	–
	Connections	–	Communication with the sales manager	–
	Competition	–	Raw material supply to competing companies	–

Note: “–” indicates no applicable sub-criterion.

3 Methodology

3.1 Analytic Network Process

The ANP extends the AHP by allowing interdependence and feedback among decision elements. The ANP was selected because sustainability criteria exhibit interdependencies that the AHP cannot capture. For example, environmental management systems (C10) influence resource consumption (C11) and pollution reduction (C13), while cost (C1) affects flexibility in giving discounts (C9). The ANP involves four steps:

Step 1: Model Construction. The decision problem is structured as a network of clusters (economic, environmental, and social) containing nodes (criteria). Arrows indicate dependencies between criteria.

Step 2: Pairwise Comparison Matrices. Experts compare criteria pairwise using Saaty's 1–9 scale. For each expert, an individual matrix is constructed. These individual matrices are aggregated using the geometric mean:

$$a_{ij}^{agg} = \left(\prod_{k=1}^K a_{ij}^k \right)^{1/K}$$

where, K is the number of experts.

Step 3: Supermatrix Formation. The aggregated pairwise comparisons form an unweighted supermatrix. This is normalized to create a weighted supermatrix, which is raised to limiting powers until convergence yields the limit supermatrix, obtaining the final criteria weights.

Step 4: Weight Extraction. Criteria weights are extracted from the limit supermatrix.

3.2 Multi-Criteria Optimization and Compromise Solution

VIKOR is a multi-criteria decision-making method for solving discrete decision problems with conflicting and incommensurable criteria. This method determines a ranked set of alternatives based on proximity to the ideal solution [24]. The main goal of the VIKOR method is to approach the ideal response for each indicator. To solve the model and obtain the matrices of pairwise comparisons, the opinions of experts in this field were considered. In the first stage, experts with relevant academic and industrial experience were interviewed about which criteria are effective in each criterion, and the effectiveness criteria were determined according to each criterion, thus avoiding unnecessary pairwise comparisons. To obtain the pairwise comparison matrices, a comparative scale was considered for experts to give their opinions. Based on their knowledge and experience, experts provided their answers with qualitative value to prepare the matrix of pairwise comparisons. Then, the experts' pairwise comparison judgments were aggregated using the geometric mean, and the final criteria weights were computed with Super Decisions software using the ANP. The steps of the VIKOR method were then executed.

Briefly, the steps of the proposed framework are as follows:

Step 1: Specify the expert(s) who has enough knowledge about the suppliers and their background.

Step 2: Determine the criteria/sub-criteria that are necessary for the sustainability goals of this issue.

Step 3: Collect the experts' opinions regarding the criteria's importance.

Step 4: Aggregate the experts' pairwise comparison judgments through the geometric mean, and determine the final weights of the criteria using Super Decisions software and the ANP.

Step 5: Construct and solve the model using the collected data in Step 3 through the VIKOR method.

Step 6: Identify the best value (f_j^+) and the worst value (f_j^-) of each criterion $j = 1, 2, \dots, n$, over the alternatives $i = 1, 2, \dots, m$. Because the direction of preference differs between benefit and cost criteria, they are defined separately:

For benefit criteria (higher is better; C4–C17):

$$f_j^+ = \max_i f_{ij}, f_j^- = \min_i f_{ij}$$

For cost criteria (lower is better; C1–C3):

$$f_j^+ = \min_i f_{ij}, f_j^- = \max_i f_{ij}$$

where, f_{ij} is the performance of alternative i on criterion j , m is the number of alternatives, and n is the number of criteria. f_j^+ is the best (positive ideal) value and f_j^- is the worst (negative ideal) value of criterion j . Consequently, for cost criteria the best alternative is the one with the lowest value.

Step 7: At this stage, the amount of S_i (desirability) and R_i (dissatisfaction index) is calculated according to the following formula. For this purpose, first, the obtained weights are multiplied in the decision matrix and the values of S_i and R_i are obtained.

$$S_i = \sum_{j=1}^n w_j (f_j^+ - f_{ij}) / (f_j^+ - f_j^-)$$

$$R_i = \max_j [w_j (f_j^+ - f_{ij}) / (f_j^+ - f_j^-)]$$

Since f_j^+ and f_j^- are defined according to the criterion type, the normalized distance $(f_j^+ - f_{ij}) / (f_j^+ - f_j^-)$ lies in $[0,1]$ for both benefit and cost criteria, with 0 indicating the best performance. Here w_j is the ANP weight of criterion j , and the sum runs over $j = 1, \dots, n$.

Step 8: Calculate the VIKOR index Q_i , which represents the compromise ranking measure where lower values indicate better alternatives.

$$Q_i = v \left[\frac{S_i - S^+}{S^- - S^+} \right] + (1 - v) \left[\frac{R_i - R^+}{R^- - R^+} \right]$$

where, v is the weight assigned to the strategy of maximum group utility (typically set to 0.5), and S^+ , S^- , R^+ , and R^- are the minimum and maximum values of S and R across alternatives.

Step 9: The alternatives are ranked by S , R and Q in ascending order, giving three ranking lists. The alternative $A^{(1)}$ with the lowest Q is proposed as the compromise solution if both conditions are satisfied:

C1 (acceptable advantage): $Q(A^{(2)}) - Q(A^{(1)}) \geq DQ$, where $DQ = 1/(m-1)$, $A^{(2)}$ is the alternative ranked second by Q , and m is the number of alternatives.

C2 (acceptable stability): $A^{(1)}$ must also be ranked best by S and/or R .

If C2 is not satisfied, $A^{(1)}$ and $A^{(2)}$ are proposed together. If C1 is not satisfied, the set $A^{(1)}, A^{(2)}, \dots, A^{(M)}$ is proposed, where $A^{(M)}$ has the largest position M satisfying $Q(A^{(M)}) - Q(A^{(1)}) < DQ$.

Figure 1 shows the methodological framework for selecting the sustainable glass suppliers, illustrating the three-phase approach. Phase 1 involves criteria selection through literature review and expert consultation. Phase 2 involves weight determination using the ANP. In Phase 3, the VIKOR method is used for supplier ranking, which is validated through sensitivity analysis and method comparison.

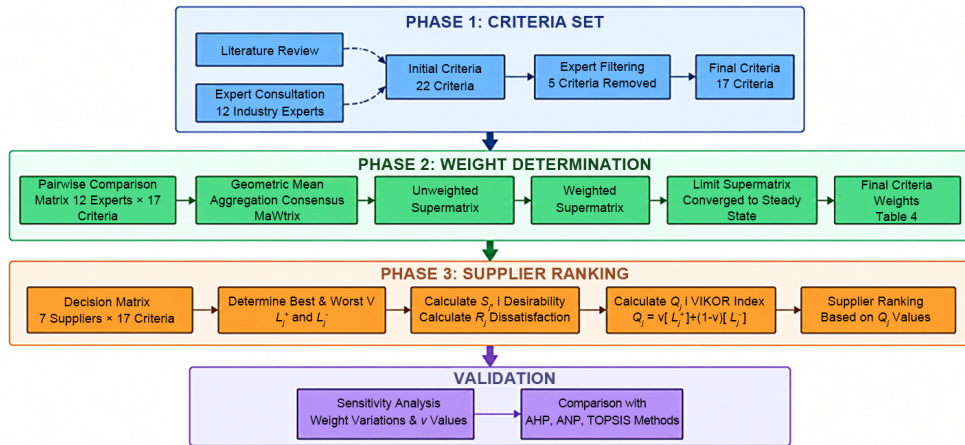


Figure 1. Methodological framework

Note: ANP = analytic network process; AHP = Analytic Hierarchy Process; TOPSIS = Technique for Order Preference by Similarity to Ideal Solution.

4 Case Study

A real case study was conducted at a major Iranian glass manufacturing company to demonstrate the framework's applicability. Twelve experts and specialists with relevant experience in supplier selection within the glass manufacturing industry participated in this research. Table 2 summarizes the expert profiles.

After finalizing criteria, expert opinions were collected through individual questionnaires and structured interviews. Each expert independently provided pairwise comparison matrices for the 17 criteria. These individual matrices were aggregated using the geometric mean to produce a consensus matrix. Seven suppliers were evaluated and measured. The suppliers (A1–A7) were selected based on market share exceeding 5%, geographic distribution across Iran, and data availability. They represent the major glass raw material suppliers operating in Iran's domestic market. Initially, 22 criteria were identified from the literature. After expert consultation, 5 criteria were removed as they were deemed less relevant to the glass industry context: geographical location was considered less critical due to Iran's centralized production; disclosure of information and company brand were deemed less influential; connections with suppliers were considered routine; and competition among suppliers was not a primary selection factor. This resulted in 17 final criteria (C1–C17), as described in Table 3.

Table 2. Expert profiles

Expert Identification	Academic Background	Work Area	Experience (Years)
E1	Doctor of Philosophy in industrial engineering	Procurement management	18
E2	Doctor of Philosophy in business administration	Supply chain management	20
E3	Master of Science in industrial engineering	Operations management	15
E4	Master of Science in quality management	Quality control	12
E5	Master of Science in environmental engineering	Environmental management	10
E6	Master of Science in logistics management	Logistics	16
E7	Master of Science in industrial engineering	Production planning	14
E8	Doctor of Philosophy in supply chain management	Strategic sourcing	22
E9	Master of Science in industrial engineering	Supplier development	9
E10	Master of Science in business administration	Corporate strategy	17
E11	Doctor of Philosophy in industrial engineering	Operations research	19
E12	Master of Science in environmental science	Sustainability	8

Note: “E” stands for Expert.

Table 3. Supplier selection criteria in this study

Dimensions	Criteria	Sub-Criteria	Index
Economic	Cost	Fixed price	C1
		After-sales service costs	C2
	Quality	Maintenance costs	C3
		Internal material quality inspection process	C4
	Delivery	Quality	C5
		Timely delivery	C6
	Services	Delay payment	C7
		After-sales service	C8
Environmental	Flexibility	After-sales service	C9
		ISO 14001 certification	C11
		Energy consumption	C12
		Dangerous gases	C13
Social	Pollution reduction	Technology	C14
		Transportation	C15
	Occupational health and safety	Provision of appropriate equipment in the workplace	C16
		Employee rights and welfare	C17

Note: “C” stands for Criterion.

The experts' pairwise comparison judgments were aggregated using the geometric mean, and the final weights of the criteria were determined using Super Decisions software and the ANP (Step 4). The supermatrix was constructed from pairwise comparisons provided by the experts. The inconsistency rate was calculated as 0.001, which is below the acceptable threshold of 0.1. The weighted supermatrix was then raised to limiting powers until convergence, yielding the limit supermatrix presented in Table 4. As shown, all columns of the limit supermatrix are identical, and the values in each row represent the final priority weight of the corresponding criterion. These weights are summarized in Table 5. The decision matrix is presented in Table 6.

After constructing the model (Step 5), the best and worst values of each criterion were obtained according to Table 7 (Step 6). Then S_i (desirability) and R_i (dissatisfaction index) values were calculated according to Table 8 (Step 7), and, the VIKOR index which is the final score of each alternative was obtained according to Table 9 (Step 8). Table 10 shows the ordered list of Q , R , and S and the ranking. Table 7 presents the best (f_j^+) and worst (f_j^-) values for each criterion. f_j^+ represents the best value (maximum for benefit criteria, minimum for cost criteria) for criterion j , while f_j^- represents the worst value. For cost criteria (C1, C2, and C3), lower values are better; for benefit criteria (C4–C17), higher values are better.

Table 4. Limit matrix resulting from the ANP

	C1	C2	C3	C4	C5	C6	C7	C8	C9	C10	C11	C12	C13	C14	C15	C16	C17
C1	0.139	0.139	0.139	0.139	0.139	0.139	0.139	0.139	0.139	0.139	0.139	0.139	0.139	0.139	0.139	0.139	0.139
C2	0.038	0.038	0.038	0.038	0.038	0.038	0.038	0.038	0.038	0.038	0.038	0.038	0.038	0.038	0.038	0.038	0.038
C3	0.026	0.026	0.026	0.026	0.026	0.026	0.026	0.026	0.026	0.026	0.026	0.026	0.026	0.026	0.026	0.026	0.026
C4	0.093	0.093	0.093	0.093	0.093	0.093	0.093	0.093	0.093	0.093	0.093	0.093	0.093	0.093	0.093	0.093	0.093
C5	0.180	0.180	0.180	0.180	0.180	0.180	0.180	0.180	0.180	0.180	0.180	0.180	0.180	0.180	0.180	0.180	0.180
C6	0.071	0.071	0.071	0.071	0.071	0.071	0.071	0.071	0.071	0.071	0.071	0.071	0.071	0.071	0.071	0.071	0.071
C7	0.035	0.035	0.035	0.035	0.035	0.035	0.035	0.035	0.035	0.035	0.035	0.035	0.035	0.035	0.035	0.035	0.035
C8	0.084	0.084	0.084	0.084	0.084	0.084	0.084	0.084	0.084	0.084	0.084	0.084	0.084	0.084	0.084	0.084	0.084
C9	0.054	0.054	0.054	0.054	0.054	0.054	0.054	0.054	0.054	0.054	0.054	0.054	0.054	0.054	0.054	0.054	0.054
C10	0.028	0.028	0.028	0.028	0.028	0.028	0.028	0.028	0.028	0.028	0.028	0.028	0.028	0.028	0.028	0.028	0.028
C11	0.072	0.072	0.072	0.072	0.072	0.072	0.072	0.072	0.072	0.072	0.072	0.072	0.072	0.072	0.072	0.072	0.072
C12	0.035	0.035	0.035	0.035	0.035	0.035	0.035	0.035	0.035	0.035	0.035	0.035	0.035	0.035	0.035	0.035	0.035
C13	0.047	0.047	0.047	0.047	0.047	0.047	0.047	0.047	0.047	0.047	0.047	0.047	0.047	0.047	0.047	0.047	0.047
C14	0.052	0.052	0.052	0.052	0.052	0.052	0.052	0.052	0.052	0.052	0.052	0.052	0.052	0.052	0.052	0.052	0.052
C15	0.023	0.023	0.023	0.023	0.023	0.023	0.023	0.023	0.023	0.023	0.023	0.023	0.023	0.023	0.023	0.023	0.023
C16	0.015	0.015	0.015	0.015	0.015	0.015	0.015	0.015	0.015	0.015	0.015	0.015	0.015	0.015	0.015	0.015	0.015
C17	0.008	0.008	0.008	0.008	0.008	0.008	0.008	0.008	0.008	0.008	0.008	0.008	0.008	0.008	0.008	0.008	0.008

Note: ANP = analytic network process; "C" stands for Criterion.

Table 5. Final criteria weights

Criterion	Criterion Description	Weight	Criterion	Criterion Description	Weight
C1	Cost fixed price	0.139	C10	ISO 14001 certification	0.028
C2	After-sales service costs	0.038	C11	Energy consumption	0.072
C3	Maintenance costs	0.026	C12	Dangerous gases	0.035
C4	Internal material quality inspection process	0.093	C13	Technology	0.047
C5		0.180	C14	Transportation	0.052
C6	Timely delivery	0.071	C15	Provision of appropriate equipment in the workplace	0.023
C7	Delay payment	0.035	C16	Employment insurance	0.015
C8	After-sales service	0.084	C17	Standard working hours	0.008
C9	Flexibility in giving discounts	0.054	Total		1.000

Note: For cost criteria (C1–C3), f_j^* is the minimum and f_j^- is the maximum value across alternatives; for benefit criteria (C4–C17), f_j^* is the maximum and f_j^- is the minimum. f_j^* denotes the best value (positive ideal solution) and f_j^- the worst value (negative ideal solution).

Table 6. Decision matrix

Criterion	Qazvin Glass Company	Ardakan Yazd Glass Company	Sahand Tabriz	Kaviyan Jam Shiraz	Kaveh	Abnoosjam	Lia Glass
C1	38000	37500	40000	38500	40000	55000	40000
C2	390	365	410	395	405	580	420
C3	775	760	840	800	830	1150	830
C4	5	4	4	4	3	3	2
C5	5	5	4	4	4	3	3
C6	1	1	1	1	0	0	0
C7	1	1	0	0	0	0	0
C8	5	5	3	4	3	2	3
C9	1	1	0	0	0	0	0
C10	1	1	1	1	1	0	1
C11	1	1	1	0	0	0	0
C12	1	1	0	0	0	0	0
C13	1	1	1	1	0	0	0
C14	5	4	4	4	2	3	3
C15	5	5	4	4	4	3	2
C16	1	1	1	1	1	0	0
C17	1	1	1	1	1	0	0

Note: C1–C3 are cost criteria (lower is better) expressed in \$. C4, C5, C8, C14 and C15 are scored on a 1–5 scale (1 = very poor, 5 = excellent). C6, C7, C9–C13, C16 and C17 are binary indicators (1 = supplier satisfies the criterion, 0 = it does not). C7 delay payment exist 1 and 0, otherwise. C12 dangerous gases exist 1 and 0, otherwise.

Table 7. Best and worst values of the decision matrix

Criterion	f_j^*	f_j^-
C1	37500	55000
C2	365	580
C3	760	1150
C4	5	2
C5	5	3
C6	1	0
C7	1	0
C8	5	2
C9	1	0
C10	1	0
C11	1	0
C12	1	0
C13	1	0
C14	5	2
C15	5	2
C16	1	0
C17	1	0

Note: C = Criterion; f_j^* denotes the best value (positive ideal solution) for criterion j ; f_j^- denotes the worst value (negative ideal solution) for criterion j .

Table 8. Calculated S and R values

Criterion/ Measure	Qazvin Glass	Ardakan Yazd Glass	Sahand Tabriz	Kaviyan Jam Shiraz	Kaveh	Abnoosjam	Lia
C1	0.004	0.000	0.020	0.008	0.020	0.139	0.020
C2	0.004	0.000	0.008	0.005	0.007	0.038	0.010
C3	0.001	0.000	0.005	0.003	0.005	0.026	0.005
C4	0.000	0.031	0.031	0.031	0.062	0.062	0.093
C5	0.000	0.000	0.090	0.090	0.090	0.180	0.180
C6	0.000	0.000	0.000	0.000	0.071	0.071	0.071
C7	0.000	0.000	0.035	0.035	0.035	0.035	0.030
C8	0.000	0.000	0.056	0.028	0.056	0.084	0.056
C9	0.000	0.000	0.054	0.054	0.054	0.054	0.050
C10	0.000	0.000	0.000	0.000	0.020	0.028	0.000
C11	0.000	0.000	0.000	0.072	0.072	0.072	0.072
C12	0.000	0.000	0.035	0.035	0.035	0.035	0.030
C13	0.000	0.000	0.000	0.000	0.047	0.047	0.047
C14	0.000	0.017	0.017	0.017	0.052	0.035	0.035
C15	0.000	0.000	0.008	0.008	0.008	0.015	0.023
C16	0.000	0.000	0.000	0.000	0.010	0.015	0.015
C17	0.000	0.000	0.000	0.000	0.008	0.008	0.008
S	0.009	0.048	0.359	0.386	0.613	0.944	0.758
R	0.004	0.031	0.090	0.090	0.090	0.180	0.180

Note: C = Criterion; S = desirability index; R = dissatisfaction index.

Table 9. VIKOR index values

Alternative	Supplier	S	R	Q
A1	Qazvin Glass Company	0.009	0.004	0.000
A2	Ardakan Yazd Glass Company	0.048	0.031	0.097
A3	Sahand Tabriz	0.359	0.090	0.431
A4	Kaviyan Jam Shiraz	0.386	0.090	0.445
A5	Kaveh	0.613	0.090	0.567
A6	Abnoosjam	0.944	0.180	1.000
A7	Lia Glass	0.758	0.180	0.900

Note: A = alternative (supplier); S = desirability index; R = dissatisfaction index; VIKOR = Višekriterijumska optimizacija i kompromisno rešenje (Multi-Criteria Optimization and Compromise Solution). Q = VIKOR compromise index. Lower values of S , R , and Q indicate better performance.

Table 10. VIKOR index ranking

Alternative	Supplier	<i>S</i>	<i>R</i>	<i>Q</i>
A1	Qazvin Glass Company	1	1	1
A2	Ardakan Yazd Glass Company	2	2	2
A3	Sahand Tabriz	3	3	3
A4	Kaviyan Jam Shiraz	4	3	4
A5	Kaveh	5	3	5
A6	Abnoosjam	7	6	7
A7	Lia Glass	6	6	6

Note: A = alternative (supplier); *S* = desirability index; *R* = dissatisfaction index; VIKOR = Višekriterijumska optimizacija i kompromisno rešenje (Multi-Criteria Optimization and Compromise Solution). *Q* = VIKOR compromise index. Lower values of *S*, *R*, and *Q* indicate better performance.

Tied *R* values receive the same rank; *Q* resolves the final ordering. With $m = 7$, $DQ = 1/6 = 0.167$. Since $Q(A2) - Q(A1) = 0.097 < DQ$, condition C1 is not satisfied. Condition C2 is satisfied because A1 is ranked first by *S*, *R* and *Q*. Because $Q(A3) - Q(A1) = 0.431 > DQ$, the compromise set consists of A1 (Qazvin Glass Company) and A2 (Ardakan Yazd Glass Company).

5 Sensitivity Analysis

Sensitivity analysis was conducted on critical variables. The first part examines the effect of different criteria weights, while the second part examines different values of v , according to Table 11. In this section, each criterion weight was changed to its maximum possible value, and the remaining criteria were assigned the minimum possible value [25]. This approach determines each alternative's strengths and weaknesses for each criterion. From the 17 sub-criteria, each criterion in turn was assigned the dominant weight of 0.392, and the remaining 16 criteria were each assigned 0.038 (so that the weights sum to 1). Alternatives were then ranked using VIKOR ($v = 0.5$).

When each criterion was considered individually as the dominant criterion, A1 and A2 held the best positions in all scenarios, whereas A6 generally showed the poorest performance (Figure 2). The x-axis lists the criteria (C1–C17) and the y-axis shows the (1-*Q*) value ranging from 0 to 1. The (1-*Q*) value represents the complement of the compromise ranking, where higher values indicate better performance.

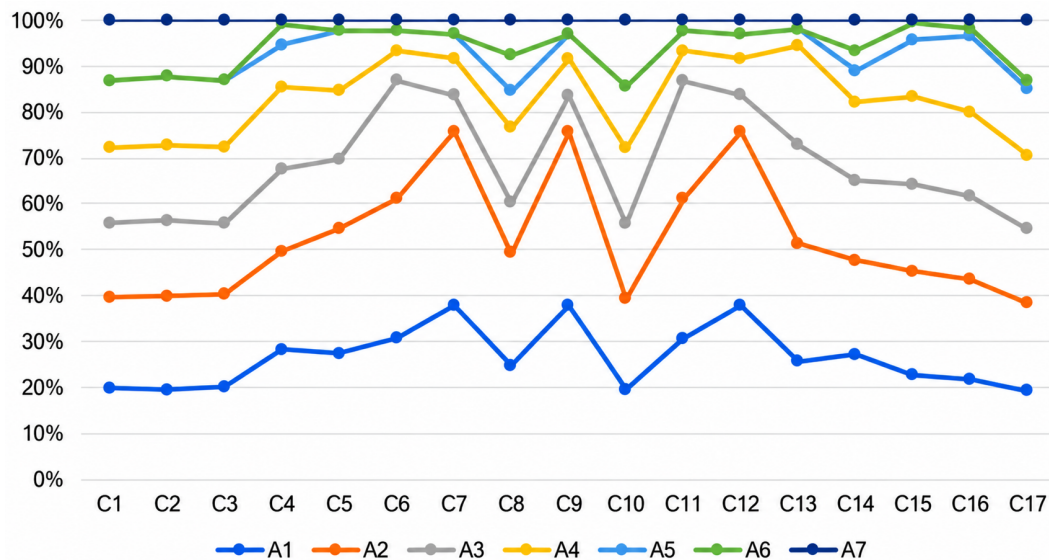


Figure 2. Sensitivity analysis of (1-*Q*) values for each of the sub-criteria

Note: A = alternative (supplier); *Q* = VIKOR compromise index; C = criteria. Higher (1-*Q*) values indicate better performance. The x-axis lists the 17 criteria (C1–C17). VIKOR = Višekriterijumska optimizacija i kompromisno rešenje (Multi-Criteria Optimization and Compromise Solution).

Table 11. Weight criteria

Scenario (Dominant Criterion)	C1	C2	C3	C4	C5	C6	C7	C8	C9	C10	C11	C12	C13	C14	C15	C16	C17
Weight	0.392	0.038	0.038	0.038	0.038	0.038	0.038	0.038	0.038	0.038	0.038	0.038	0.038	0.038	0.038	0.038	0.038
A1	38000	390	775	5	5	1	1	5	1	1	1	1	1	5	5	1	1
A2	37500	365	760	4	5	1	1	5	1	1	1	1	1	4	5	1	1
A3	40000	410	840	4	4	1	0	3	0	1	1	0	1	4	4	1	1
A4	38500	395	800	4	4	1	0	4	0	1	0	0	1	4	4	1	1
A5	40000	405	830	3	4	0	0	3	0	1	0	0	0	2	4	1	1
A6	55000	580	1150	3	3	0	0	2	0	0	0	0	0	3	3	0	0
A7	40000	420	830	2	3	0	0	3	0	1	0	0	0	3	2	0	0

Note: A = alternative (supplier); C = Criterion.

5.1 Sensitivity Analysis of the v Value

The coefficient v represents the importance of each component, typically set to 0.5. Lower v values emphasize individual opinions (such as veto rights in decision-making), while higher v values emphasize collective opinions. For the sensitivity analysis, v was varied from 0 to 1. The previous ranking was based on $v = 0.5$. After calculating Q values for different values of v ranging from 0 to 1, the results were obtained (Table 12).

Table 12. Q values of the alternatives for different values of v

	$v = 0$	$v = 0.1$	$v = 0.2$	$v = 0.3$	$v = 0.4$	$v = 0.5$	$v = 0.6$	$v = 0.7$	$v = 0.8$	$v = 0.9$	$v = 1$
A1	0	0	0	0	0	0	0	0	0	0	0
A2	0.151	0.140	0.129	0.118	0.108	0.097	0.086	0.075	0.064	0.053	0.042
A3	0.487	0.476	0.465	0.453	0.442	0.431	0.420	0.408	0.397	0.386	0.374
A4	0.487	0.479	0.471	0.462	0.454	0.445	0.437	0.428	0.420	0.411	0.403
A5	0.487	0.503	0.519	0.535	0.551	0.567	0.583	0.599	0.614	0.630	0.646
A6	1	1	1	1	1	1	1	1	1	1	1
A7	1	0.980	0.960	0.940	0.920	0.900	0.881	0.861	0.841	0.821	0.801

Note: A = alternative (supplier); v = weight of the strategy of maximum group utility, ranging from 0 to 1; lower v emphasizes individual regret, higher v emphasizes group utility; $v = 0.5$ represents a balanced compromise.

Table 13. (1- Q) values of the alternatives

	$v = 0$	$v = 0.1$	$v = 0.2$	$v = 0.3$	$v = 0.4$	$v = 0.5$	$v = 0.6$	$v = 0.7$	$v = 0.8$	$v = 0.9$	$v = 1$
A1	1	1	1	1	1	1	1	1	1	1	1
A2	0.849	0.860	0.871	0.882	0.892	0.903	0.914	0.925	0.936	0.947	0.958
A3	0.513	0.524	0.535	0.547	0.558	0.569	0.580	0.592	0.603	0.614	0.626
A4	0.513	0.521	0.529	0.538	0.546	0.555	0.563	0.572	0.580	0.589	0.597
A5	0.513	0.497	0.481	0.465	0.449	0.433	0.417	0.401	0.386	0.370	0.354
A6	0	0	0	0	0	0	0	0	0	0	0
A7	0	0.020	0.040	0.060	0.080	0.100	0.119	0.139	0.159	0.179	0.199

Note: A = alternative (supplier); v = weight of the strategy of maximum group utility, ranging from 0 to 1; lower v emphasizes individual regret, higher v emphasizes group utility; $v = 0.5$ represents a balanced compromise.

To better show the priority of the alternatives, the values of (1- Q) are shown in Table 13. The (1- Q) value represents the complement of the compromise ranking, where higher values indicate better performance. The change of values according to different v is shown in Figure 3.

In Figure 3, the x-axis lists the alternatives (A1–A7), the y-axis shows the (1- Q) value ranging from 0 to 1, and each series corresponds to a value of v from 0 to 1 in steps of 0.1. Higher (1- Q) values indicate better performance.

The overall ranking does not change for $v > 0$, and A1 and A2 remain the top alternatives; at $v = 0$, A3, A4 and A5 are tied ($Q = 0.487$). A1 and A6 are unaffected by v . A2 approaches the ideal solution as v increases and consensus importance rises. Only A5 moves away from the ideal solution as v increases.

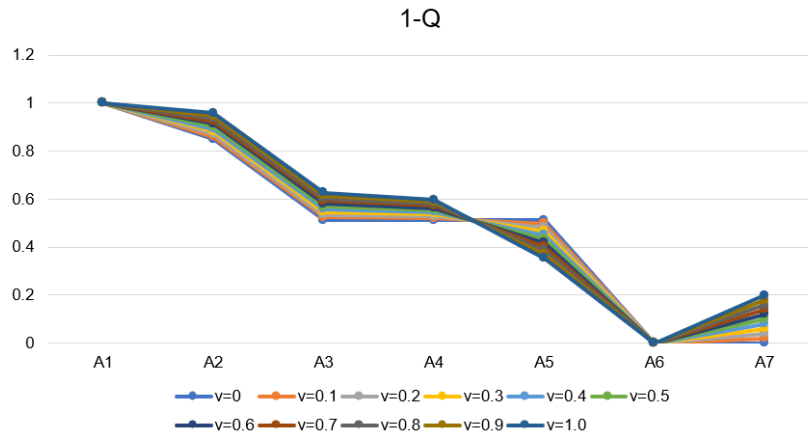


Figure 3. Chart of (1-Q) values of the alternatives in terms of different v values

Note: A = alternative (supplier); v = weight of the strategy of maximum group utility, ranging from 0 to 1; lower v emphasizes individual regret, higher v emphasizes group utility; $v = 0.5$ represents a balanced compromise.

6 Discussion

The VIKOR method results were compared with those obtained using the AHP, ANP, and TOPSIS methods. After obtaining criteria weights via the ANP, alternatives were ranked using all four methods. The standardized values of the methods used are shown in Table 14.

Table 14. Values of different methods

	VIKOR	AHP	ANP	TOPSIS
A1	1	1	1	1
A2	0.903	0.892	0.971	0.988
A3	0.569	0.525	0.527	0.600
A4	0.555	0.533	0.416	0.426
A5	0.433	0.237	0.178	0.130
A6	0	0	0	0
A7	0.100	0.020	0.095	0.071

Note: VIKOR = Višekriterijumska optimizacija i kompromisno rešenje (Multi-Criteria Optimization and Compromise Solution); AHP = Analytic Hierarchy Process; ANP = Analytic Network Process; TOPSIS = Technique for Order Preference by Similarity to Ideal Solution; A = alternative (supplier); Q = VIKOR compromise index.

The process of changing the scores of alternatives across different calculation methods is shown in Figure 4.

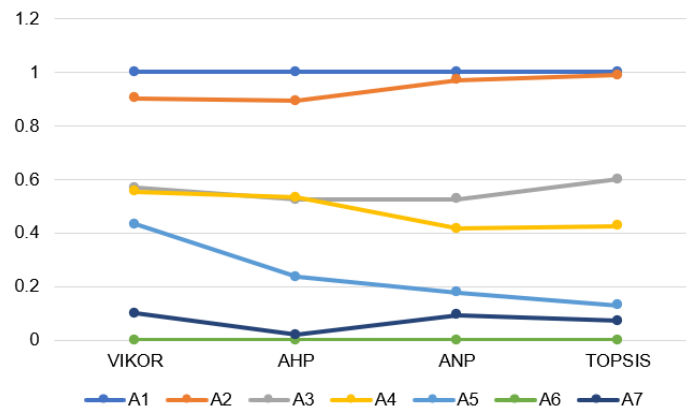


Figure 4. Changes in the priority of alternatives across different calculation methods

Note: VIKOR = Multi-Criteria Optimization and Compromise Solution; AHP = Analytic Hierarchy Process; ANP = Analytic Network Process; TOPSIS = Technique for Order Preference by Similarity to Ideal Solution; A = alternative (supplier).

6.1 Comparative Findings

The ranking consistency across methods confirms the robustness of the proposed integrated ANP-VIKOR approach. All methods identified Qazvin Glass Company and Ardakan Yazd Glass Company as the top two suppliers, suggesting reliable prioritization. AHP reversed the order of A3 and A4, whereas VIKOR, ANP and TOPSIS agreed on the full ranking [24]. In short, the results and ranking of the alternatives in the above methods can be displayed as follows. The ranking of all methods in determining the best alternative was the same. The ranking of alternatives in different methods is shown in Table 15.

Table 15. Prioritizing alternatives based on different methods

	VIKOR	AHP	ANP	TOPSIS
A1	1	1	1	1
A2	2	2	2	2
A3	3	4	3	3
A4	4	3	4	4
A5	5	5	5	5
A6	7	7	7	7
A7	6	6	6	6

Note: VIKOR = Višekriterijumska optimizacija i kompromisno rešenje (Multi-Criteria Optimization and Compromise Solution); AHP = Analytic Hierarchy Process; ANP = Analytic Network Process; TOPSIS = Technique for Order Preference by Similarity to Ideal Solution; A = alternative (supplier).

6.2 Comparison with Recent Studies

The findings of this study align with those of the study by Rodrigues et al. [10], who observed that economic criteria dominated while social criteria remained “practically inert” in a glass packaging company case. This pattern suggests industry-specific sustainability dynamics where economic performance considerations often overshadow social dimensions. Similarly, Afrasiabi et al. [9] identified pollution control and environmental management systems as most influential criteria in manufacturing contexts. In the present study, environmental criteria carried a combined weight of 0.234 but individually ranked below quality and cost, suggesting a partial divergence attributable to the glass industry context. However, the results of the current study contrast with those of the study by Memari et al. [11], who found social criteria more influential in automotive spare parts manufacturing. This divergence may reflect industry-specific characteristics, where automotive supply chains face greater regulatory scrutiny regarding labor practices and social compliance. The glass industry’s raw material extraction and energy-intensive production processes may naturally prioritize environmental considerations over social considerations. The proposed ANP captured interdependencies among sustainability criteria, a capability highlighted by Chen et al. [14] as essential for smart supply chain applications. The weight distribution shows that quality (0.180) and cost (0.139) remain paramount, consistent with traditional supplier selection criteria [1]. Environmental criteria carried a combined weight of 0.234; within them, energy consumption (0.072) and transportation (0.052) received the highest weights, while ISO 14001 certification received 0.028. Social criteria received minimal weights (C15–C17: 0.008–0.023; cluster total 0.046), confirming the social inertia phenomenon identified by Rodrigues [19].

6.3 Managerial Implications

Based on the findings, several practical actions can be recommended for managers:

- **Supplier quality performance.** With quality receiving the highest weight (0.180), managers should establish formal supplier quality audit programs. This includes developing standardized quality scorecards, conducting regular on-site inspections, and implementing quality performance tracking systems. Specific metrics should include defect rates, compliance with specifications, and consistency of raw material quality.
- **Cost-value analysis.** Given cost’s second-highest weight (0.139), managers should implement comprehensive total-cost-of-ownership models that capture fixed price, maintenance costs, and after-sales service costs. This requires collecting and analyzing historical cost data, establishing cost benchmarks, and creating supplier cost-performance dashboards for decision-making.
- **Environmental compliance.** With increasing environmental regulation, managers should incorporate environmental criteria (ISO 14001 certification, energy consumption, dangerous gas emissions, technology, and transportation) into supplier selection scorecards. This involves requesting environmental documentation, conducting environmental audits, and setting minimum environmental performance thresholds.
- **Social sustainability.** Given the minimal influence of social criteria (0.008–0.023), managers should develop phased approaches to social sustainability integration. This includes requiring occupational health and

safety documentation, verifying employment insurance compliance, and ensuring standard working hours. These requirements can be introduced progressively over 12–24 months to allow suppliers to adapt.

- Implementation of the integrated ANP-VIKOR framework. The proposed framework provides systematic supplier evaluation. Managers should formalize this as a standard decision process for supplier selection, including (a) establishing a cross-functional evaluation team, (b) collecting performance data across all criteria, (c) applying the ANP-VIKOR methodology using available software (e.g., Super Decisions), and (d) using results for periodic supplier review and development.

- Adaptation to glass-industry requirements. The framework should be tailored to glass industry specifics: prioritization of uninterrupted supply for continuous furnace operation, inclusion of raw material quality consistency metrics, and consideration of transportation logistics optimization.

6.4 Social Dimension Examination

The social dimension analysis revealed minimal influence on supplier selection decisions (weights ranging from 0.008 to 0.023). This finding has important implications for practical decision-making. In the Iranian glass industry context, social criteria (occupational health and safety, employment insurance, and standard working hours) currently have limited impact on supplier selection. This suggests that buyers primarily evaluate suppliers on economic and quality performance. While social criteria currently receive limited weight, managers should recognize this as a potential vulnerability. As international trade and sustainability regulations evolve, social criteria may become more prominent. Managers should therefore consider gradually incorporating social criteria into supplier evaluation scorecards, strengthening supplier capacity for social compliance, monitoring emerging regulatory requirements in export markets, and promoting greater awareness of corporate social responsibility among suppliers. Social sustainability should be addressed through: (a) basic certification requirements (health and safety, and labor compliance), (b) periodic social audits, (c) supplier development programs for social capacity building, and (d) inclusion of social performance in long-term supplier relationship management.

7 Conclusions, Limitations, and Future Research

The VIKOR method, combined with the ANP for weight determination, provided an effective framework for sustainable glass supplier selection. This study identified criteria affecting sustainable glass supply chain selection, extracted sub-criteria from internal/external sources and expert interviews, and incorporated them into standard clusters. Questionnaires captured sub-criteria relationships and weights, with Super Decisions software determining criteria weights. The VIKOR method subsequently selected the optimal alternatives. The findings imply that quality (0.180) and cost (0.139) emerged as the most influential criteria⁵; Qazvin Glass Company and Ardakan Yazd Glass Company were consistently identified as optimal suppliers; social criteria received minimal weights, confirming industry-specific sustainability dynamics; and the ANP successfully captured interdependencies among sustainability criteria. Sensitivity analysis confirmed result robustness across criteria weight variations and v values. The findings provide glass industry decision-makers with a systematic framework for sustainable supplier selection. The limited influence of social criteria suggests opportunities for industry-wide initiatives to elevate social sustainability awareness. The integrated ANP-VIKOR approach offers a practical tool applicable to other industrial sectors.

Several limitations should be acknowledged. The case study focuses exclusively on the Iranian glass industry, which may limit generalizability to other countries or regions. Future research should examine other geographic contexts with different regulatory and market conditions. The research relied on 12 experts from a single industry sector. A broader expert sample, including diverse perspectives, could enhance generalizability. In addition, the criteria were derived from literature and expert consultation specific to the glass industry. Different industries or contexts may require different criteria sets. Sustainability priorities may change over time with evolving regulations and market expectations. Longitudinal studies are needed to examine how criteria weights evolve. Furthermore, the ANP-VIKOR approach assumes consistent expert judgments and rational decision-making. Future research could explore behavioral aspects of supplier selection decisions. The limited weight of social criteria may reflect the specific context rather than universal patterns. Future research should examine factors that might increase social criteria importance.

Future research could validate the framework across different industries and geographical contexts using larger and more diverse expert samples. Longitudinal studies could also examine changes in sustainability priorities and criterion weights over time. Methodologically, fuzzy extensions of VIKOR, including spherical, Fermatean, and picture fuzzy approaches, could be investigated to better represent uncertainty in expert judgments. The incorporation of behavioral factors and managerial constraints may further enhance the practical applicability of sustainable supplier selection models.

Author Contributions

Conceptualization, S.B.; methodology, S.B.; software, S.B.; validation, H.F.; formal analysis, H.F.; investigation, N.G.; data curation, H.F.; writing—original draft preparation, H.F.; writing—review and editing, N.G.; visualization, N.G. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data supporting our research results are included within the article.

Conflicts of Interest

The authors declare no conflicts of interest.

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